

CONNECTED PSEUDOSPECTRA OF A NONNORMAL TOEPLITZ PATH: THE OPEN HATANO–NELSON CHAIN*

SHANMU JIN[†]

Abstract. For the nonnormal tridiagonal Toeplitz matrix $A_n(a) = \text{tridiag}(a, 0, 1)$, $0 < a < 1$, we determine the finite-size transition at which its Euclidean ε -pseudospectrum becomes connected. The least singular value of $(x + iy)I_n - A_n(a)$ is nondecreasing in $|y|$, so every pseudospectral component retracts onto an interval of the real axis and is therefore contractible. If γ_n is the largest least-singular-value barrier between adjacent open-boundary eigenvalues, then connectedness is equivalent to $\varepsilon > \gamma_n$. We prove the strict finite-size comparison $\gamma_{n+1} < \gamma_n$ and sharp-order bounds $\gamma_n \asymp_a (n+1)a^{n/2}$. Consequently, the first connected matrix size is also the eventual connectedness threshold, and, for fixed a , it equals $2(\log(1/\varepsilon) + \log\log(1/\varepsilon))/\log(1/a) + O_a(1)$ as ε tends to zero. In the open-boundary Hatano–Nelson model, the coefficient $2/\log(1/a)$ is the right-eigenmode amplitude skin depth, in lattice-spacing units; the threshold describes merger of worst-case spectral-uncertainty clouds at a prescribed perturbation level.

Key words. pseudospectrum, nonnormal Toeplitz matrix, Hatano–Nelson model, non-Hermitian skin effect, least singular value, Chebyshev polynomial

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1. Introduction. For a matrix $A \in \mathbb{C}^{n \times n}$, the Euclidean ε -pseudospectrum is

$$(1.1) \quad \sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}.$$

Equivalently, it is the union of the spectra of $A + \Delta A$ over all unstructured complex perturbations $\Delta A \in \mathbb{C}^{n \times n}$ with $\|\Delta A\|_2 < \varepsilon$; see [26]. For nonnormal matrices, these sets reveal spectral sensitivity and resolvent growth that the eigenvalues alone can miss; see [24, 26]. Classical spectral limit sets for banded Toeplitz sections go back to Schmidt and Spitzer [22]. Finite Toeplitz matrices are a setting in which the distinction between spectra and pseudospectra can be exponentially large in the dimension; their finite-section resolvents and pseudospectra have been studied in [6, 7, 8, 21].

We study the real tridiagonal Toeplitz path

$$(1.2) \quad A_n(a) = \text{tridiag}(a, 0, 1) = \begin{bmatrix} 0 & 1 & & \\ a & 0 & \ddots & \\ & \ddots & \ddots & 1 \\ & & a & 0 \end{bmatrix}, \quad n \geq 2, \quad 0 < a < 1,$$

where the three arguments of tridiag denote the subdiagonal, diagonal, and super-diagonal, respectively. Closed-form spectral, conditioning, and pseudospectral results for general tridiagonal Toeplitz matrices are developed in [17]. Although $A_n(a)$ is diagonally similar to a symmetric Jacobi matrix, the similarity is not unitary, and hence it does not preserve the Euclidean pseudospectrum. Our question is finite-dimensional and topological: as n grows at fixed a and ε , can connectedness appear and then disappear, or is there a unique transition?

The same family is the clean open-boundary Hatano–Nelson chain [12, 13]. With the site convention in which t_R moves a particle from j to $j+1$ and t_L moves it from

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[†]Department of Neurosurgery, Peking Union Medical College Hospital, No. 1 Shuaifuyuan, Dongcheng District, Beijing 100730, China (jinshanmu1129@pku.edu.cn).

38 $j + 1$ to j , assume $0 < t_R < t_L$ and write

$$39 \quad H_n^{\text{OBC}} = t_L A_n(a), \quad a = \frac{t_R}{t_L}.$$

40 The dimensionless logarithmic parametrization used in the proof is

$$41 \quad (1.3) \quad h = -\frac{1}{2} \log a > 0, \quad a = e^{-2h}, \quad r = \sqrt{a} = e^{-h}.$$

42 The diagonal gauge

$$43 \quad \mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1})$$

44 and the upper-shift matrix $V_n = \text{tridiag}(0, 0, 1)$ give the dimensionless similarity

$$45 \quad (1.4) \quad A_n(a) = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1}.$$

46 Multiplying by t_L and using $t_L r = \sqrt{t_L t_R}$ recovers the usual physical imaginary-gauge
 47 formula for H_n^{OBC} . Thus the open-boundary eigenvalues are real, while the right
 48 eigenmodes have the envelope $e^{-h(j-1)}$ and accumulate at the left edge. This is the
 49 non-Hermitian skin effect [18, 27, 28]; see [29] for a review. Under periodic boundary
 50 conditions the Bloch curve is the ellipse

$$51 \quad \mathcal{E}_{\text{PBC}}(k) = t_L e^{ik} + t_R e^{-ik} = (t_L + t_R) \cos k + i(t_L - t_R) \sin k.$$

52 Early work connecting Hatano–Nelson–type localization with pseudospectra treated
 53 random bidiagonal and Toeplitz models [25]. Direct pseudospectral interpretations
 54 of nonnormal boundary sensitivity appear in [19, 20]. Mathematical foundations for
 55 the skin effect were developed in [2], while spectra and pseudospectra of tridiagonal k -
 56 Toeplitz matrices were treated in [4, 5]. Pseudospectral scaling in asymmetric lattices
 57 was studied in [15]; see also the preprint [23] for a closely related pseudospectral
 58 analysis of the open Hatano–Nelson chain.

59 For general matrices, the geometry and coalescence of pseudospectral components
 60 have been studied by variational and distance-to-defectivity methods [1, 9, 16]. The
 61 present work complements that theory by determining, for the clean scalar tridiagonal
 62 Toeplitz path, an exact finite-dimensional connectedness criterion, contractibility
 63 of every component, and a strict comparison of the merger barrier in consecutive
 64 dimensions.

65 Our contributions are as follows. First, we prove vertical monotonicity of the
 66 least singular value, which reduces planar topology to a real-axis barrier problem and
 67 shows, more strongly, that every pseudospectral component is contractible. Second, a
 68 signed persymmetric pencil and a folded Chebyshev parametrization give the strict
 69 comparison of the largest barrier in consecutive dimensions. Third, rectangular
 70 factorizations give sharp-order two-sided bounds and the resulting Lambert- W finite-
 71 size law. Connectedness concerns the union over all *unstructured* perturbations: it does
 72 not imply that any single perturbation supplies eigenvalues throughout a connecting
 73 bridge, nor does it describe the attainable set for a prescribed structured perturbation
 74 family. The transition is also not the onset of the skin effect, which is already present
 75 in (1.4).

76 The paper is organized as follows. Section 2 states the main result. Vertical
 77 monotonicity is proved in section 3; the strict barrier comparison occupies section 4;
 78 and the topological criterion and quantitative bounds are established in section 5.
 79 Numerical illustrations and the physical interpretation are given in sections 6 and 7.

2. Model and main results. For $n \geq 2$ and $0 < a < 1$, let

$$A_n(a) = \text{tridiag}(a, 0, 1).$$

When a is fixed, we write A_n for $A_n(a)$. For $\varepsilon > 0$, write

$$\sigma_\varepsilon(A_n) = \{z \in \mathbb{C} : s_{\min}(zI_n - A_n) < \varepsilon\}.$$

Define

$$N_e(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_n) \text{ is connected for every } n \geq N\},$$

$$N_f(a, \varepsilon) = \min\{N \geq 2 : \sigma_\varepsilon(A_N) \text{ is connected}\}.$$

Throughout the remainder of the article, put

$$r = \sqrt{a}, \quad J_n = [\mathbf{1}_{\{i+j=n+1\}}]_{i,j=1}^n.$$

Thus $A_n = V_n + aV_n^T$. Throughout, $\mathbf{e}_j$ denotes a Euclidean coordinate vector of the ambient dimension, and $U_j(t)$ denotes the Chebyshev polynomial of the second kind, with $U_{-1} = 0$.

When the dimension is n , write $N = n + 1$ and define

$$(2.1) \quad I_n^{\text{sp}} = \left[-2r \cos \frac{\pi}{N}, 2r \cos \frac{\pi}{N}\right], \quad g_n(x) = s_{\min}(xI_n - A_n), \quad \gamma_n = \max_{x \in I_n^{\text{sp}}} g_n(x).$$

The explicit lower and upper barriers used below are

$$(2.2) \quad \mathcal{U}_n(a) = r^n \csc \frac{\pi}{N},$$

$$(2.3) \quad \mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \frac{r^n(1 - r^2)}{(1 - r^{2N}) \sin \frac{3\pi}{2N}}.$$

THEOREM 2.1 (Main theorem). *For every $0 < a < 1$ and $\varepsilon > 0$, the following assertions hold.*

1. *For every $n \geq 2$, every connected component of $\sigma_\varepsilon(A_n(a))$ is contractible.*
2. *For $n \geq 2$, the barrier sequence is strictly decreasing, $\gamma_{n+1} < \gamma_n$, tends to zero, and*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n.$$

3. *The first-hit and eventual connectedness thresholds coincide:*

$$N_c(a, \varepsilon) := N_f(a, \varepsilon) = N_e(a, \varepsilon).$$

Moreover,

$$(2.4) \quad 1 + \max\{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\} \leq N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\},$$

where the maximum of the empty set is understood to be 1.

4. *For fixed a , as $\varepsilon \downarrow 0$,*

$$(2.5) \quad N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1),$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$.

3. Vertical monotonicity. We first prove the vertical monotonicity that will be used in the topological part of the argument.

LEMMA 3.1 (Vertical monotonicity). *For every real x ,*

$$y \mapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\mathcal{P}_n(Y, t) = ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \quad F_n(Y, t) = \det \mathcal{P}_n(Y, t).$$

The same definition is used in every dimension: for $k \geq 1$, $\mathcal{P}_k(Y, t)$ is obtained by replacing n by k , $F_k(Y, t) = \det \mathcal{P}_k(Y, t)$, and $F_0 = 1$. We shall prove the finite-dimensional implication

$$(3.1) \quad \mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

At a fixed value of Y , write

$$d = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$d - 1, d, \dots, d, d - a^2,$$

whereas for $n = 1$ its sole entry is $d - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal d and these two off-diagonals. Set $T_{-1} = 0$, $T_0 = 1$, and put

$$(3.2) \quad D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}.$$

Indeed, a determinant is affine in each diagonal entry. Replacing the first diagonal entry d by $d - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $d - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

We record the continuant calculation that contains all the dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$(3.3) \quad \begin{aligned} T_k &= dT_{k-1} - (|b|^2 - a^2)T_{k-2} - (2a^2d - a(b^2 + \bar{b}^2))T_{k-3} \\ &\quad - a^2(|b|^2 - a^2)T_{k-4} + a^4dT_{k-5} - a^6T_{k-6} - a^2\mathbf{1}_{\{k=2\}}. \end{aligned}$$

For example, $T_1 = d$ and $T_2 = d^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (3.3). Multiplying the recurrence by z^k and summing gives

$$(3.4) \quad \sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)},$$

where

$$\begin{aligned} Q(z, Y) &= 1 - dz + (|b|^2 - a^2)z^2 + (2a^2d - a(b^2 + \bar{b}^2))z^3 \\ &\quad + a^2(|b|^2 - a^2)z^4 - a^4dz^5 + a^6z^6. \end{aligned}$$

Consequently, with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$,

$$(3.5) \quad \mathcal{D}(z, Y) = \frac{(1-z)(1-a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1-a^2 z) \mathcal{D}(z, Y).$$

For the second identity, first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of d and b gives

$$\partial_Y Q = -z(1-z)(1+az)^2(1-a^2 z).$$

Differentiating (3.5) therefore yields

$$(3.6) \quad \sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2 z)^2}{1-az}.$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$(3.7) \quad D_k > 0, \quad R_k > 0 \quad (0 \leq k < n).$$

Put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. The coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$(3.8) \quad \begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ &\quad + 2(1-a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned}$$

Only the third coefficient can be negative. We now give the cofactor inequality that absorbs it. Let $j, k \geq 1$, $j+k = n-1$. Deleting index $k+1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , coupled only by the second off-diagonal a . In block form, the deleted principal submatrix is

$$(3.9) \quad \begin{bmatrix} L_k & a\mathbf{e}_k \mathbf{e}_1^T \\ a\mathbf{e}_1 \mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$(3.9) \quad D_k R_j - a^2 D_{k-1} R_{j-1} > 0.$$

Deleting index $j+1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (3.2) gives

$$(3.10) \quad R_k = D_k + (1-a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}.$$

Hence $\vartheta_k \leq a^2$ implies $\varpi_k \leq 1$. On dividing (3.9) and its companion by positive factors, we obtain

$$(3.11) \quad \vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2.$$

If both ϑ_j and ϑ_k were at most a^2 , then both ϖ_j and ϖ_k would be at most 1, contradicting (3.11). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1 - a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (3.11) now gives $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$(3.12) \quad \vartheta_j \vartheta_k > a^4.$$

Consequently, for $n \geq 3$, applying (3.12) to $j = p + 1$, $k = n - 2 - p$ gives

$$D_{p+1} D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$(3.13) \quad u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}.$$

If the third coefficient in (3.8) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (3.8) and (3.13) prove (3.1). If that coefficient is nonnegative, the implication is immediate. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (3.1) holds in every dimension.

Finally fix $Y_0 \geq 0$. If the least singular value at Y_0 is zero, the desired inequality is automatic. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (3.1) says that its determinant is strictly increasing. By continuity of Hermitian eigenvalues, a finite right endpoint would be a first loss of positive definiteness at a positive-semidefinite singular matrix, and hence would have determinant zero. This is impossible because the determinant has increased from its positive value at Y_0 . Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

4. Strict finite-size comparison of real-gap barriers. We next establish the strict comparison between consecutive dimensions. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity gives existence of every maximum, and the spectral endpoints have height zero. Hence

$$(4.1) \quad \gamma_n = \max_{1 \leq k < n} \gamma_{n,k}.$$

LEMMA 4.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

We give the complete finite-path calculation. Persymmetry gives $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$(4.2) \quad s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}.$$

Let $q_n(x, s) = \det(xI_n - A_n - sJ_n)$.

LEMMA 4.2 (The middle signed branch). *Write the eigenvalues of the real symmetric matrix*

$$B_n(x) = (xI_n - A_n)J_n$$

in decreasing order as $\alpha_1(x) \geq \dots \geq \alpha_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \alpha_m(x), & n = 2m, \\ \alpha_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. Since $A_n = \mathcal{G}_n(r)A_n^{\text{sym}}\mathcal{G}_n(r)^{-1}$ and $\mathcal{G}_n(r)^{-1}J_n = r^{-(n-1)}J_n\mathcal{G}_n(r)$,

$$(4.3) \quad B_n(x) = r^{-(n-1)}\mathcal{G}_n(r)(xI_n - A_n^{\text{sym}})J_n\mathcal{G}_n(r).$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (4.3) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

Counting signs gives

$$(4.4) \quad \nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m - 1, & j \text{ odd}, \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m + 1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases}$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m - 1, \dots, m + 1).$$

In these coordinates

$$(4.5) \quad \begin{aligned} B_{2m}(x) &= \widehat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \widehat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned}$$

where $P_m(x) = xI_m - A_m$. Thus $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $\Gamma = \text{diag}(I_m, -I_m)$,

$$\tilde{B} = -\Gamma B_{2m}(x)\Gamma = \hat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If $\tilde{\alpha}_1 \geq \cdots \geq \tilde{\alpha}_{2m}$ are the eigenvalues of $\tilde{B}$, then

$$(4.6) \quad \tilde{\alpha}_k = -\alpha_{2m+1-k}.$$

Positive-semidefinite rank-two interlacing gives

$$(4.7) \quad \tilde{\alpha}_k \geq \alpha_k \geq \tilde{\alpha}_{k+2}, \quad 1 \leq k \leq 2m-2.$$

If j is even, the only candidates for minimum modulus have moduli α_m and $-\alpha_{m+1} = \tilde{\alpha}_m$, and $\alpha_m \leq \tilde{\alpha}_m$. If j is odd and $m \geq 2$, the two candidates have moduli α_{m-1} and $-\alpha_m = \tilde{\alpha}_{m+1}$; taking $k = m-1$ in (4.7) gives $\alpha_{m-1} \geq \tilde{\alpha}_{m+1}$. For $m = 1$, both eigenvalues in the sole gap are negative, so the larger one α_1 has minimum modulus. Thus α_m always attains the minimum modulus, and its sign follows from (4.4). Equality in an interlacing comparison is harmless: it only gives a multiple least singular value.

Now let $n = 2m+1$. Group the first and last m coordinates as above and leave the middle coordinate last. Then

$$(4.8) \quad B_{2m+1}(x) = \begin{bmatrix} \hat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}.$$

In an open spectral gap of A_{2m+1} , $P_m(x)$ is nonsingular: indeed, its zeros are the even-indexed eigenvalues of A_{2m+1} . The eigenvalues of the principal submatrix $\hat{P}_m(x)$ are consequently

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\alpha_m \geq s_m(P_m) \geq \alpha_{m+1} \geq -s_m(P_m) \geq \alpha_{m+2}.$$

If j is even, (4.4) and interlacing give

$$0 < \alpha_{m+1} \leq s_m(P_m) \leq -\alpha_{m+2};$$

if j is odd, they give

$$0 < -\alpha_{m+1} \leq s_m(P_m) \leq \alpha_m.$$

Every other eigenvalue is farther from zero. Hence α_{m+1} attains the minimum modulus, with sign prescribed by (4.4); equality again only means multiplicity. At a spectral endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. Finally, multiplication by J_n does not change singular values, which proves the assertion. $\square$

Proof of Lemma 4.1. Folded continuants. Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2 + (1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\begin{aligned} \Delta_{\text{fold}} &= ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2), \\ \mathfrak{b} &= (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned}$$

The global recurrence polynomial is
(4.9)

$$\begin{aligned} \Pi(z) &:= 1 - C_0z + \mathfrak{b}z^2 - a^2C_0z^3 + a^4z^4, \\ \left. \begin{aligned} 2a(u+v) &= C_0, \\ a^2(4uv+2) &= \mathfrak{b} \end{aligned} \right\} \implies \Pi(z) &= (1 - 2auz + a^2z^2)(1 - 2avz + a^2z^2), \quad u, v \in \mathbb{R}. \end{aligned}$$

On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair $(u, v) = (\xi_+, \xi_-)$ satisfies the two symmetric relations in (4.9):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

The proof uses the root presentation only on these real sheets; the global transfer and generating identities use the polynomial Π itself. Here is a direct continuant derivation of the required folded formulas. Pair the coordinates in the order $(1, n), (2, n-1), \dots$. In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^{\text{e}} = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix};$$

for odd order the terminal block is the scalar $x-s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $\mathcal{Q}_m^{\text{e}} = q_{2m}(x, s)$, $\mathcal{Q}_m^{\text{o}} = q_{2m+1}(x, s)$. We include the continuant verification to fix all boundary signs. We now give a complete minor transfer for the folded continuant. For $\tau \in \{\text{e}, \text{o}\}$, let K_m^{τ} denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m+1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^{e} has order $2m$, K_m^{o} has order $2m+1$, and

$$K_1^{\text{e}} = \begin{bmatrix} -1-s & x \\ x & -a-s \end{bmatrix}, \quad K_1^{\text{o}} = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x-s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$(4.10) \quad K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = \begin{bmatrix} E_{\text{off}} & 0 \end{bmatrix},$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

Here is the one-step calculation. On the dense set on which both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of K_m in (4.10) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \tag{4.11}$$

Both sides are polynomial in x and s , so these identities hold also when one of the Schur complements is singular.

The five quantities in (4.11) satisfy one linear invariant. Define

$$\ell_m = (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m.$$

Direct substitution into (4.11) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1 + s)(a + s) - x^2, -a - s, -1 - s, x, 1), \\ &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1 + a^2) + 2as - C_0(x - s), \\ &\quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s). \end{aligned}$$

Thus $\ell_1 = 0$ in both parity classes, and consequently

$$(4.12) \quad (1-a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1-a)x\mathfrak{c}_m - a(1-a)\mathfrak{r}_m = 0 \quad (m \geq 1).$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first signed minor below is precisely the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m [\mathfrak{d}_m^\tau \quad \mathfrak{p}_m^\tau \quad \mathfrak{q}_m^\tau \quad \mathfrak{c}_m^\tau]^T, \quad (\boldsymbol{\eta}_m^e)_1 = \mathcal{Q}_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = \mathcal{Q}_m^o.$$

Eliminating $\mathfrak{r}_m$ with (4.12) reduces (4.11) to the four-dimensional transfer

$$(4.13) \quad \boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}.$$

The terminal conditions can be incorporated at $m = 0$ by setting

$$(4.14) \quad \boldsymbol{\eta}_0^e = [1 \quad a^{-1} \quad 1 \quad 0]^T, \quad \boldsymbol{\eta}_0^o = [x-s \quad 1 \quad 1 \quad -1]^T.$$

Indeed, direct multiplication in (4.13) gives

$$\mathsf{T}_{\text{fold}} \boldsymbol{\eta}_0^e = \begin{bmatrix} x^2 - (1+s)(a+s) \\ a+s \\ 1+s \\ -x \end{bmatrix},$$

and

$$\mathsf{T}_{\text{fold}} \boldsymbol{\eta}_0^o = \begin{bmatrix} C_0(x-s) - x(1+a^2) - 2as \\ a^2 - s^2 + sx \\ 1 - s^2 + sx \\ a + sx - x^2 \end{bmatrix},$$

which are exactly the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$(4.15) \quad \det(\lambda I_4 - \mathsf{T}_{\text{fold}}) = \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

Cayley–Hamilton, followed by multiplication by $\mathsf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$ and extraction of the first coordinate, therefore yields, for $Z_m = \mathcal{Q}_m^e$ and for $Z_m = \mathcal{Q}_m^o$,

$$(4.16) \quad Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4.$$

Finally, applying T_{fold} three times to (4.14) gives the boundary residuals

$$\text{Res}_0 = Z_0, \quad \text{Res}_1 = Z_1 - C_0 Z_0, \quad \text{Res}_2 = Z_2 - C_0 Z_1 + \mathfrak{b} Z_0,$$

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$$\text{Res}_3 = Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0,$$

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namely

	Res ₀	Res ₁	Res ₂	Res ₃
$Z_m = \mathcal{Q}_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = \mathcal{Q}_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

as asserted. Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$. This proves (4.16). Since $\Pi(0) = 1$, it is invertible in $\mathbb{R}[[z]]$, and the two generating functions are the literal formal-power-series identities

$$(4.17) \quad \sum_{m \geq 0} \mathcal{Q}_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1},$$

and

$$(4.18) \quad \sum_{m \geq 0} \mathcal{Q}_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}.$$

Now let $U_{-1} = 0$. The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u)) \omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$\sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}}) U_m(u) \omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

For a differentiable real function F , write

$$[F]_{u,v} := \begin{cases} \frac{F(u) - F(v)}{u - v}, & u \neq v, \\ F'(u), & u = v. \end{cases}$$

Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (4.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers (4.17) and (4.18). Coefficient comparison gives

$$(4.19) \quad q_{2m}(x, s) = a^m [\mathbf{f}_m]_{u,v}, \quad \mathbf{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)),$$

and

$$(4.20) \quad q_{2m+1}(x, s) = a^m [\mathbf{g}_m]_{u,v}, \quad \mathbf{g}_m(t) = \left((x - s)t - \frac{x(1 + a^2) + 2as}{2a} \right) U_m(t).$$

The determinant recurrence and generating identities are global. The real-pair formulas (4.19)–(4.20) are used only after the two symmetric relations have been verified; their

collision value is the displayed derivative. No complex square-root choice is made when $\Delta_{\text{fold}} < 0$, and none is needed below.

Half-angle chord parametrization. Recall that $a = e^{-2h}$, and put

$$\mathfrak{s}_h = \sinh h, \quad \mathfrak{c}_h = \cosh h.$$

For $K \geq 2$ and $0 \leq \theta \leq \pi$, define

$$(4.21) \quad \Phi_K(\theta) = \sqrt{\mathfrak{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta},$$

where the quotient is interpreted continuously at $\theta = 0, \pi$, and

$$\Psi_K(\eta) = \sqrt{\mathfrak{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h,$$

with the limiting value at $\eta = 0$. We spell out the half-angle reduction, including its sign. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathfrak{p}_\theta = \sqrt{\mathfrak{c}_h^2 - y^2}, \quad \mathfrak{p}_\eta = \sqrt{\mathfrak{c}_h^2 - z^2},$$

and let $L_0 = 2r/\mathfrak{c}_h$. If $x = L_0 y z$, $s = L_0 \mathfrak{p}_\theta \mathfrak{p}_\eta > 0$, and $\tau = \varsigma s$ with $\varsigma \in \{-1, 1\}$, then direct expansion gives

$$(4.22) \quad \begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathfrak{c}_h^2 - 1 + 2\varsigma \mathfrak{p}_\theta \mathfrak{p}_\eta. \end{aligned}$$

The real pair (u, v) satisfies the two symmetric relations in (4.9). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (4.19) and (4.20), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$(4.23) \quad \mathfrak{p}_\theta U_{K-1}(y) = \varsigma \mathfrak{p}_\eta U_{K-1}(z).$$

In the odd-fold case the reduction also contains the factor $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$. On a distinct-variable sheet this factor cannot vanish: after squaring, its vanishing would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\text{sgn}(U_{K-1}(\cos \theta)) = \text{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (4.23) is equivalent to $\varsigma = (-1)^j$ and

$$(4.24) \quad \Phi_K(\theta) = \Psi_K(\eta),$$

with

$$(4.25) \quad s = \mathfrak{s}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta.$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the preceding steps are reversible, and the two chord relations are equivalent to the corresponding signed equation. The determinant identities themselves extend to a folded-variable collision by the continuous divided-difference convention, but we invoke the chord equivalence only for distinct variables; the selected middle branch is proved below to be collision-free. Here $K-1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$, $0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $\mathfrak{s}_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathfrak{s}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

Branch classification and exhaustion. We next identify exactly which of these chords is the middle branch of Lemma 4.2. Set

$$(4.26) \quad \mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h.$$

At $u = \pm \mathfrak{c}_h$, this denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$(4.27) \quad - \sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0.$$

At each finite nodal endpoint the logarithmic derivative tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow \mathfrak{c}_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a j -chord is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term endpoint-side always refers to the part of the outer lobe adjacent to $\mathfrak{c}_h$, not to the hyperbolic or elliptic coordinate used for it.

We now prove that these chords are exactly the middle signed branch. By reflection it suffices to consider a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{F}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$(4.28) \quad \Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u).$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and $\mathfrak{c}_h$. Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k} I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

Continue this ordered middle root through the open gap, and let $\mathcal{I}$ be the maximal subinterval on which

$$\theta \in (j\pi/K, (j+1)\pi/K), \quad z = z_K(\mathcal{W}_K(\cos \theta)).$$

The ordered eigenvalues of the continuous real symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so a multiple least singular value does not interrupt this continuation. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ is real and continuous by construction. It can collide only when $z^2 = y^2$, which is impossible because

$$z > u_K^{\max} > \cos \frac{\pi}{K} > \cos \theta = y.$$

Thus no endpoint of $\mathcal{I}$ can arise from a folded-variable collision inside the open gap. The signs of the half variables cannot change in a noncentral positive gap: $z > 0$ throughout, while $y = 0$ would force $x = 0$, outside such a gap. Moreover, z cannot reach $\mathfrak{c}_h$ in the interior, because (4.25) would then give $s = 0$. The inner variable cannot reach a nodal endpoint: the common level would be zero, the endpoint-side solution would be $\mathfrak{c}_h$, and (4.25) would give $s = 0$, whereas the middle root has no zero inside a spectral gap. The outer variable cannot leave its endpoint-side component, since its first such point would be $u_K^{\max}$, contradicting (4.28). Passage through $z = 1$ is harmless; only its coordinate changes. Therefore $\mathcal{I}$ is the entire gap.

At the two endpoints the inner parameter tends to $j\pi/K$ and $(j+1)\pi/K$. Continuity and the intermediate value theorem show that every parameter in that nodal interval occurs. Conversely, for a fixed inner parameter the endpoint-side outer solution

is unique, so every j -chord is a point of the ordered middle branch. By Lemma 4.2, its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$. This proves both continuation and exhaustion without assuming that $\theta \mapsto x$ is monotone.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the identity $x = (2r/\mathfrak{c}_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half.

Noncentral gap comparison. We now compare the noncentral middle branches. Let $N \geq 3$, put $M = N + 1$, and set $\kappa = N/M$. Every positive noncentral gap of the M -problem satisfies

$$(4.29) \quad j \geq 1, \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

Indeed, such gaps satisfy $j+1 \leq N/2$. Hence $\theta_0 = \theta/\kappa$ belongs to the same-indexed N -nodal interval, and every trigonometric argument below lies in $(0, \pi/2]$. Consider a point of such a branch for the matrix of order $M - 1$, with

$$(4.30) \quad \theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right), \quad \frac{(j+1)\pi}{N} \leq \frac{\pi}{2}.$$

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Put $\theta_0 = \theta/\kappa$. Then θ_0 belongs to the corresponding N -nodal interval. Concavity of sine gives

$$(4.31) \quad \frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{F}_h(\theta_0)}{\mathcal{F}_h(\theta)} > \kappa.$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$(4.32) \quad \mathcal{F}_h(\theta_0)^2 - \kappa^2 \mathcal{F}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$(4.33) \quad \frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa,$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Hence $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\widehat{\theta}$ of

$$(4.34) \quad \Phi_N(\widehat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\widehat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$(4.35) \quad \frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$(4.36) \quad \frac{d}{dt} (t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

508 The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The
 509 displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$
 510 follows by continuity. The chord classification above says that the constructed value
 511 $\mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (4.25) and
 512 the strict increase of sine on the interval now give

$$513 \quad (4.32) \quad \mathfrak{s}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathfrak{s}_{\text{hyp}}(\theta, \eta).$$

514 Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$,
 515 $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$516 \quad \theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

517 The ratio

$$518 \quad \mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{F}_h(t/\kappa)}{\mathcal{F}_h(t)}$$

519 is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$520 \quad \frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

521 because

$$522 \quad t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

523 is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second
 524 decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The
 525 rightmost solution of

$$526 \quad \Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

527 therefore has $\widehat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) =$
 528 $(\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and
 529 differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$ gives

$$530 \quad \kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

531 Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-
 532 side condition for the N -outer solution. The one-maximum property now fixes the
 533 label, and the chord classification says again that $\mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0)$ is the least singular value
 534 at the constructed abscissa. Here $\widehat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below
 535 $\pi/2$. Thus both sine factors in the elliptic version of (4.25) have increased, at least
 536 one strictly, and hence

$$537 \quad (4.33) \quad \mathfrak{s}_{\text{ell}}(\widehat{\theta}, \phi_0) > \mathfrak{s}_{\text{ell}}(\theta, \phi).$$

538 *Central gaps.* It remains to compare the central sheets. We do this in the next
 539 calculation.

540 For $m \geq 1$, let

$$541 \quad (4.34) \quad c_m = s_{\min}(aI_m + V_m).$$

By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same height; denote it by

$$(4.35) \quad d_m = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}).$$

We claim

$$(4.36) \quad c_m > d_m > c_{m+1}.$$

Put

$$(4.37) \quad \zeta = \frac{a + a^{-1}}{2}.$$

For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$(4.37) \quad aU_j(\rho_j) = U_{j-1}(\rho_j).$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$(4.38) \quad \frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$(4.38) \quad \det((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j) = a^j (U_j(t) - a^{-1}U_{j-1}(t)).$$

Thus the largest root ρ_j gives the least eigenvalue. Together with

$$(4.39) \quad U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (4.37) and (4.38) give

$$(4.39) \quad c_j^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}.$$

We shall also use that c_m is exactly the height of the central gap of A_{2m} . A permutation separating odd and even coordinates gives

$$(4.40) \quad A_{2m} \sim \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(4.41) \quad (I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are the union of the singular values of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq s_{\min}(aI_m + V_m) = c_m$, and hence $s_{\min}(A_{2m}) = c_m$. To prove that this is the maximum throughout the central gap, put $K = 2m + 1$ and use the half variables

$$(4.42) \quad y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$(4.40) \quad \mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h} yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2} (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2),$$

where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution. The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (4.37) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (4.39), this yields

$$(4.41) \quad \mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = \mathfrak{c}_m^2.$$

At $x = 0$, Lemma 4.2 and the chord-exhaustion argument identify the nearest root with c_m , so z_0 is the endpoint-side outer solution of level $\mathfrak{c}_h$. If $x \neq 0$, the level in (4.40) is strictly below $\mathfrak{c}_h$, and hence its endpoint-side solution satisfies $z > z_0$. Equations (4.40)–(4.41) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = \mathfrak{c}_m^2.$$

Reflection handles the negative half of the gap. Thus its height is exactly c_m , attained only at $x = 0$.

We first prove $d_m < c_m$. Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

Here are the details of the central specialization. Take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2\mathfrak{c}_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\mathcal{W}_{2m+2}(y)^2 = (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \quad \mathcal{W}_{2m+2}(z)^2 = (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2.$$

Because $\zeta + 1 = 2\mathfrak{c}_h^2$, the formulas $x = (2r/\mathfrak{c}_h)yz$ and $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$ become the following chord parametrization:

$$(4.42) \quad \begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) &= |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned}$$

where t_{out} is the descending, or largest, solution. To see this from (4.24), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, the descending root satisfies $t_{\text{out}} \geq \rho$, because ρ lies at or before the unique lobe maximum and the descending root lies at or after it. Suppose instead that $(\log \mathcal{V}_m)'(\rho) < 0$. Then $\rho > a$. Indeed, if $\rho \leq a < 1$, the identity

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1-\rho^2)} - \frac{1}{2(\zeta - \rho)} \\ &\geq \frac{1}{2(1-\rho)} - \frac{1}{2(\zeta - \rho)} \geq 0, \end{aligned}$$

a contradiction. The case $\rho = 1$ is immediate because $a < 1$. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = c_m^2.$$

This holds along the whole central sheet, so $d_m < c_m$.

We now prove $d_m > c_{m+1}$. Put $L = m+1 \geq 2$, and abbreviate

$$\rho = \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L,$$

$$(4.43) \quad \mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}.$$

Thus $a = u_-/u_+$ and $c_L = 1/u_+$. We first establish the scalar inequality

$$(4.44) \quad (1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L.$$

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$(4.45) \quad \mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b.$$

Indeed, $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$, so

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

and the formula for Δ_ξ follows from (4.43). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1+x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \frac{L(b-\rho)-1}{\rho^2-1}$$

with the continuous value at $\rho = 1$, and

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1+\rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (4.44) reduces it to

$$(4.46) \quad \mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0.$$

We shall repeatedly use

$$(4.47) \quad \Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1.$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

This is strictly increasing in $u \geq 1/L$, and its value at $1/L$ is positive because $\Delta_\omega L < 1$. Thus $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+(s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (4.46) holds. The case $\rho = 1$ follows by continuity.

Suppose next that $\rho < 1$. Set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Writing $\rho = \cos \theta_\rho$ gives $0 < L\theta_\rho < \pi/2$ and

$$u = b - \rho = \frac{\sin((L+1)\theta_\rho)}{\sin(L\theta_\rho)} - \cos \theta_\rho = \sin \theta_\rho \cot(L\theta_\rho).$$

Because $b > 1$, $u = \Delta_b + v > v > 0$. Also $\tan(L\theta_\rho) > L \tan \theta_\rho > L \sin \theta_\rho$, since $x \mapsto \tan x/x$ is strictly increasing on $(0, \pi/2)$. Hence

$$(4.48) \quad 0 < v < u < 1/L.$$

With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion of (4.46) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_-[2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega[u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

This is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2-v} [(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega)].$$

The elementary bounds, the second following from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L + 1)^2] < 5/(2L + 1)^2$,

$$(4.49) \quad \Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L + 1)^2} \leq \frac{1}{5}.$$

The last inequality is an equality when $L = 2$; because the preceding bound for v is strict, one still has $v < 1/5$. These bounds show that the bracketed expression is positive, since it equals

$$(2 - v)[(2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L] + 2\Delta_\omega^2(2L - 1)$$

and, using $L \geq 2$,

$$(2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (4.48),

$$\mathfrak{q}(v) = v[(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v)] > 0,$$

$$\mathfrak{q}(1/L) = v[(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L] > 0.$$

Indeed, in these two lines the positive leading terms exceed

$$\frac{9}{5} \left(4 - \frac{5}{8}\right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5}\right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$, respectively. Concavity now gives $\mathfrak{q}(u) > 0$, proving (4.46), and hence (4.44), in all cases.

It remains to certify that the scalar comparison concerns the least singular value. Define

$$(4.50) \quad x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad c = c_L.$$

At $x = x_*$ and $s = \pm c$, the real pair $u = \xi$, $v = -\omega_L$ satisfies the two symmetric relations in (4.9). Indeed, its sum and product are $\xi - \omega_L$ and $-\omega_L \xi$, and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Moreover, $\xi + \omega_L > 0$, so the distinct branch of the divided difference in (4.20) applies. For convenience, write the folded numerator as

$$\mathbf{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$\mathbf{X}_\xi = x_*(\zeta - \xi), \quad \mathbf{Y}_\xi = c(1 + \xi), \quad \mathbf{X}_\omega = x_*(\zeta + \omega_L), \quad \mathbf{Y}_\omega = c\Delta_\omega,$$

and put $U_\xi = U_{L-1}(\xi)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give $\mathbf{Y}_\omega \mathbf{Y}_\xi = \mathbf{X}_\omega \mathbf{X}_\xi$. Also

$$\frac{\mathbf{X}_\omega^2}{\mathbf{Y}_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathbf{r}_* = \mathbf{X}_\xi/\mathbf{Y}_\xi = \mathbf{Y}_\omega/\mathbf{X}_\omega \in (0, 1)$, the common nonzero factor in (4.20) is $a^{L-1}/(\xi + \omega_L)$, and the two numerator differences are

$$\begin{aligned} \mathbf{n}_c(\xi) - \mathbf{n}_c(-\omega_L) &= (1 + \mathbf{r}_*)(\mathbf{X}_\omega U_{-\omega} - \mathbf{Y}_\xi U_\xi), \\ \mathbf{n}_{-c}(\xi) - \mathbf{n}_{-c}(-\omega_L) &= (1 - \mathbf{r}_*)(\mathbf{X}_\omega U_{-\omega} + \mathbf{Y}_\xi U_\xi). \end{aligned}$$

Their product is positive by (4.44), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad \mathbf{X}_\omega^2 U_{-\omega}^2 = \frac{c^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > \mathbf{Y}_\xi^2 U_\xi^2.$$

Thus

$$(4.51) \quad q_{2L-1}(x_*, c)q_{2L-1}(x_*, -c) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. We claim

$$(4.52) \quad x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $c = a/u_-$ and $c^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$, so

$$x_* < 2\sqrt{a} \sin \theta_L, \quad c < 2\sqrt{a} \sin(\theta_L - \Delta_L).$$

It remains to prove

$$(4.53) \quad \sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L).$$

Now

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L.$$

Thus (4.53) follows from $\Delta_L \cos \theta_L > \theta_L^3$, or $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$, using $\pi^2 < 10$,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently $x_* + c < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1 - a)^2 \geq 0,$$

which proves (4.52).

Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1}) J_{2L-1}.$$

Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$q_{2L-1}(x_*, c) q_{2L-1}(x_*, -c) = \prod_{i=1}^{2L-1} (\mu_i^2 - c^2).$$

The positive product in (4.51) is nonzero and therefore says that the number of indices with $|\mu_i| < c$ is even. On the other hand, Weyl's singular-value perturbation inequality and (4.52) give, for the second-smallest singular value s_{next} ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > c,$$

so at most one index can satisfy $|\mu_i| < c$. The number is thus zero, and nonvanishing of the product excludes equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > c_L$. Since $0 < x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $d_{L-1} > c_L$. This completes the proof of (4.36).

Completion of the comparison. We finish the index comparison. Put $M = N + 1$, so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral comparison to a positive gap. Choose a point $x_j \in G_{M-1,j}$ at which its height is attained. The hyperbolic or elliptic construction above produces a point $\hat{x}_j \in G_{N-1,j}$, and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\hat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

The reflected inequality handles the corresponding negative gap.

If $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix is c_{ℓ_*} , whereas each central height of the smaller odd-order matrix is $d_{\ell_*-1} > c_{\ell_*}$. If $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $d_{\ell_*} < c_{\ell_*}$, where c_{ℓ_*} is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. Choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

5. Topology, quantitative bounds, and the transition scale. We begin with the standard component property that reduces the planar topology to the real spectral interval.

LEMMA 5.1. *Every connected component of $\sigma_\varepsilon(A_n)$ contains an eigenvalue of A_n ; this standard component property holds for arbitrary matrices [9].*

Proof. The estimate

$$s_{\min}(zI_n - A_n) \geq |z| - \|A_n\|_2$$

shows that the pseudospectrum is bounded. Let $\mathcal{C}$ be one of its components. If an eigenvalue λ belonged to $\bar{\mathcal{C}}$, continuity of the least singular value would give a ball about λ contained in the pseudospectrum. That ball meets $\mathcal{C}$, so maximality of connected components places λ in $\mathcal{C}$.

Suppose that $\mathcal{C}$ contains no eigenvalue. Its compact closure is then disjoint from the spectrum. On a neighbourhood of $\bar{\mathcal{C}}$, the function $\mathbf{u}(z) = \|(zI_n - A_n)^{-1}\|_2$ is continuous and subharmonic; see [26]. Moreover, $\mathbf{u} \leq \varepsilon^{-1}$ on $\partial\mathcal{C}$, since a strict inequality would put a neighbourhood of that boundary point in the same component. The maximum principle gives $\mathbf{u} \leq \varepsilon^{-1}$ throughout $\mathcal{C}$, contrary to the definition of the pseudospectrum. $\square$

The diagonal gauge introduced in (1.4) gives

$$A_n = \mathcal{G}_n(r) [r(V_n + V_n^T)] \mathcal{G}_n(r)^{-1},$$

and hence

$$(5.1) \quad \sigma(A_n) = \left\{ 2r \cos \frac{k\pi}{n+1} : 1 \leq k \leq n \right\}.$$

Put

$$\Omega = \sigma_\varepsilon(A_n), \quad \Omega_{\mathbb{R}} = \Omega \cap \mathbb{R} = \{x \in \mathbb{R} : g_n(x) < \varepsilon\}.$$

Because A_n is real, the least singular value is invariant under complex conjugation. Lemma 3.1 therefore implies that

$$\mathcal{H}_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

maps Ω into itself. If $\mathcal{C}$ is a component of Ω , the path $t \mapsto \mathcal{H}_t(z)$ starts in $\mathcal{C}$ and hence remains there. Thus $\mathcal{H}_t$ retracts $\mathcal{C}$ onto $\mathcal{C} \cap \mathbb{R}$.

By Lemma 5.1, this real intersection is nonempty. It is connected, being the continuous image $\mathcal{H}_1(\mathcal{C})$, and hence lies in one interval component $\mathcal{K}$ of $\Omega_{\mathbb{R}}$. Conversely, $\mathcal{K}$ is a connected subset of Ω meeting $\mathcal{C}$, so $\mathcal{K} \subset \mathcal{C}$. Therefore $\mathcal{C} \cap \mathbb{R} = \mathcal{K}$, and the components of Ω correspond bijectively to the interval components of $\Omega_{\mathbb{R}}$. In particular, each $\mathcal{C}$ retracts onto an interval and is contractible.

If Ω is connected, its real intersection is an interval containing every eigenvalue, and therefore contains their convex hull I_n^{sp} . Conversely, if $I_n^{\text{sp}} \subset \Omega$, that interval belongs to one component and contains every eigenvalue; Lemma 5.1 rules out any other component. Compactness of I_n^{sp} now gives

$$(5.2) \quad \Omega \text{ is connected} \iff I_n^{\text{sp}} \subset \Omega \iff \max_{x \in I_n^{\text{sp}}} g_n(x) < \varepsilon \iff \gamma_n < \varepsilon.$$

The strict inequality reflects the open definition of the pseudospectrum; equality leaves at least one real bridge point outside.

We now give quantitative estimates. Put $N = n + 1$ and recall $\mathcal{L}_n(a)$ and $\mathcal{U}_n(a)$ from (2.3)–(2.2).

PROPOSITION 5.2 (Sharp-order gap bounds). *For every $n \geq 2$,*

$$(5.3) \quad \mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a).$$

In particular, both bounds have order $(n + 1)a^{n/2}$, with constants depending only on a .

Proof. For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

All entries of $(xI_n - A_n)\mathbf{v}$, except the last, vanish, and the last one is $r^n \sin(N\theta)$. Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since $\pi/N \leq \theta \leq (N - 1)\pi/N$, this proves the upper bound.

For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$. For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

For $1 \leq k \leq n$, direct multiplication gives

$$(\mathbf{D}_\theta^* \mathbf{C}_\theta \mathbf{u})_k = (\mathbf{C}_\theta \mathbf{u})_k - q_\theta (\mathbf{C}_\theta \mathbf{u})_{k-1} = (p_\theta + q_\theta) u_k - u_{k+1} - p_\theta q_\theta u_{k-1}.$$

Hence

$$(5.4) \quad xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta.$$

826 The two Gram matrices are diagonally unitarily similar to the tridiagonal Toeplitz
 827 matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$,
 828 $1 \leq k \leq n$, so

$$829 \quad (5.5) \quad \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2}.$$

830 Put

$$831 \quad \mathcal{S}_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

832 Unit normal vectors to the two hyperplane ranges are

$$833 \quad \mathbf{u}_C = \mathcal{S}_N^{-1/2}(\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = \mathcal{S}_N^{-1/2}(1, q_\theta, \dots, q_\theta^n)^T.$$

834 In the polar decompositions

$$835 \quad \mathbf{C}_\theta = Q_C(\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D(\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

836 the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$
 837 intersect in dimension $n - 1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n - 1$ singular
 838 values equal to 1; the remaining one is the cosine of the remaining principal angle,
 839 $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$. The geometric sum gives

$$840 \quad |\langle \mathbf{u}_D, \mathbf{u}_C \rangle| = \frac{r^n}{\mathcal{S}_N} \left| \sum_{j=0}^{N-1} e^{i(2j-(N-1))\theta} \right|$$

$$841 \quad (5.6) \quad = \frac{r^n(1 - r^2)}{1 - r^{2N}} \left| \frac{\sin(N\theta)}{\sin \theta} \right| =: \chi_N(\theta).$$

842 Finally,

$$843 \quad xI_n - A_n = (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2} Q_D^* Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}.$$

844 Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (5.5) gives

$$845 \quad g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

846 Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

847 Finally,

$$848 \quad \mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

849 while

$$850 \quad (5.7) \quad \mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}.$$

851 These follow from $\sin \frac{\pi}{N} \geq 2/N$, $1 + r^2 - 2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$.
 852 This proves the asserted sharp order. $\square$

853 *Proof of Theorem 2.1.* By Proposition 5.2, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Lemma 4.1 says that
 854 γ_n is strictly decreasing. Equation (5.2) therefore shows that, once the pseudospectrum
 855 is connected, it stays connected in every larger dimension. This proves $N_e = N_f$.
 856 Contractibility of every component was proved immediately before (5.2).

If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum in dimension n is disconnected.

If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (2.4).

The scale in (2.4) can be displayed in closed form. Put

$$\Theta_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N \geq 2 : \Theta_m < t \text{ for every } m \geq N\}.$$

Equations (5.7) and $\mathcal{U}_n(a) \leq \Theta_n/2$ give

$$(5.8) \quad \boxed{\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon).}$$

Indeed, (5.2) gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\}.$$

The lower estimate $C_{\text{lb}}(a)\Theta_m \leq \gamma_m$ forces $\Theta_m < \varepsilon/C_{\text{lb}}(a)$ throughout the tail beginning at N_c . Conversely, $\Theta_m < 2\varepsilon$ for every $m \geq \mathcal{T}_r(2\varepsilon)$, together with $\gamma_m \leq \Theta_m/2$, forces connectedness throughout that tail. Thus the polynomial factor $n+1$, as well as the exponential factor $a^{n/2}$, is sharp on both sides.

For completeness, let

$$\beta = -\log r = \frac{1}{2} \log(1/a).$$

For $0 < t < 3r^2 = \Theta_2$, let $x_t > 2$ be the solution on the decreasing branch of $(x+1)e^{-\beta x} = t$. This branch exists because

$$0 < \beta r t < 3\beta r^3 \leq e^{-1};$$

the last inequality is $ue^{-u} \leq e^{-1}$ with $u = 3\beta$. Moreover, $t < \Theta_2$ places the descending root strictly beyond $x = 2$. Putting $y_t = x_t + 1$, the equation becomes $(-\beta y_t)e^{-\beta y_t} = -\beta r t$, whence $y_t = -\beta^{-1}W_{-1}(-\beta r t)$. Therefore

$$(5.9) \quad \mathcal{T}_r(t) = \left\lfloor -\frac{1}{\beta} W_{-1}(-\beta r t) \right\rfloor,$$

where W_{-1} is the lower real branch of the Lambert W -function. Indeed, the continuous function $(x+1)r^x$ is unimodal and, after its descending root, is strictly below t . Hence the integer tail condition is $N > x_t$; its least solution is $\lfloor x_t \rfloor + 1 = \lfloor x_t + 1 \rfloor$, including when x_t is an integer.

The standard lower-branch expansion [11]

$$-W_{-1}(-u) = \log(1/u) + \log \log(1/u) + O(1), \quad u \downarrow 0,$$

combined with (5.8) proves, for each fixed $a \in (0, 1)$, as $\varepsilon \downarrow 0$,

$$(5.10) \quad \boxed{N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log \log(1/\varepsilon)) + O_a(1).}$$

The remainder is not asserted to be uniform as $a \uparrow 1$. Finally, for $|x| \leq r$,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$. At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$. Hence, by (5.2),

$$(5.11) \quad N_c(a, \varepsilon) = 2 \iff \varepsilon > a.$$

This proves all assertions. $\square$

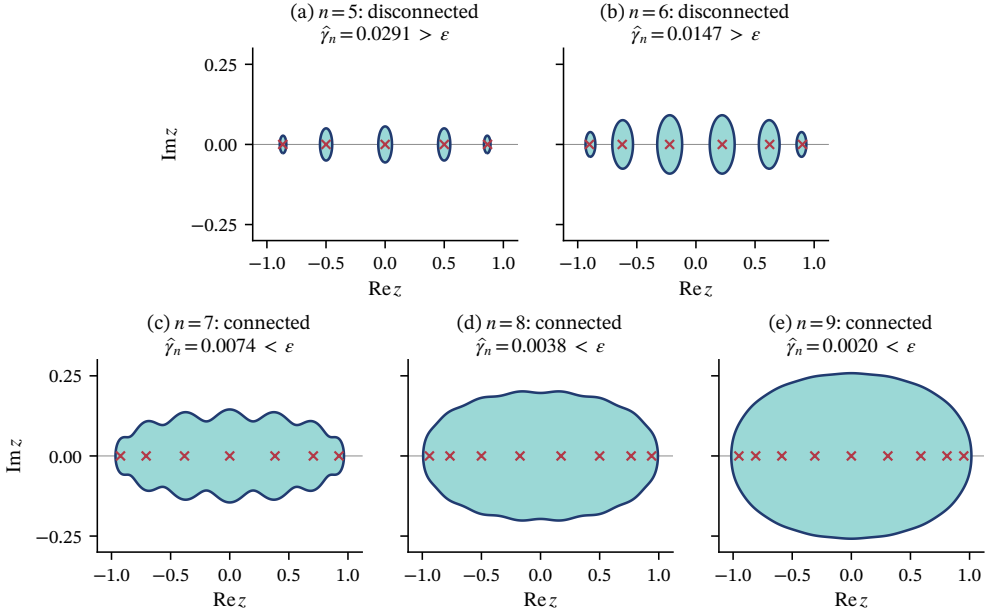


FIG. 1. Topology change at $a = 1/4$ and $\varepsilon = 10^{-2}$. Shading is $\{z : s_{\min}(zI_n - A_n) < \varepsilon\}$, and crosses are open-boundary eigenvalues. Gapwise SVD estimates give $(\hat{\gamma}_5, \dots, \hat{\gamma}_9) = (0.02909836, 0.01465829, 0.007364219, 0.003820033, 0.001992747)$. Independent Lipschitz enclosures certify the two labels above ε and the three below it, so $N_c = 7$. The merger is a perturbation-level-dependent pseudospectral event, not the onset of skin localization.

6. Numerical illustrations and reproducibility. The computations in this section illustrate the theorem but are not used in its proof. The complete script `make_hatano_nelson_figures.py`, supplied with the manuscript, uses NumPy, SciPy, and Matplotlib. It constructs $A_n(a)$ directly, computes Cartesian pseudospectral contours by singular value decomposition, and obtains gapwise estimates of γ_n by bounded scalar optimization. Independently, the 1-Lipschitz property of $x \mapsto s_{\min}(xI_n - A_n)$ gives certified lower and upper bounds from uniform real-gap grids. Thus the topology labels do not depend on pixel connectivity or an unverified optimizer output.

The five panels of Figure 1 show two disconnected cases, the first connected case, and two subsequent connected cases. In agreement with the deformation retraction, each connected cloud has no hole: every vertical section is an interval, and the real trace contains the full open-boundary spectral segment.

Figure 2 separates the two roles of the analysis. The strict comparison theorem identifies a unique crossing of a fixed ε , while the elementary upper and lower envelopes certify its sharp exponential-polynomial scale without numerically resolving the maximizing gap.

Figure 3 displays the two complementary manifestations of the imaginary gauge: the periodic/open spectral contrast and the open-boundary skin profile. Although all open-boundary eigenvalues are real, the displayed pseudospectral contours elongate rapidly toward the periodic ellipse. The ellipse is a reference curve, not a large-level limit: for sufficiently large perturbation levels the pseudospectra extend beyond it.

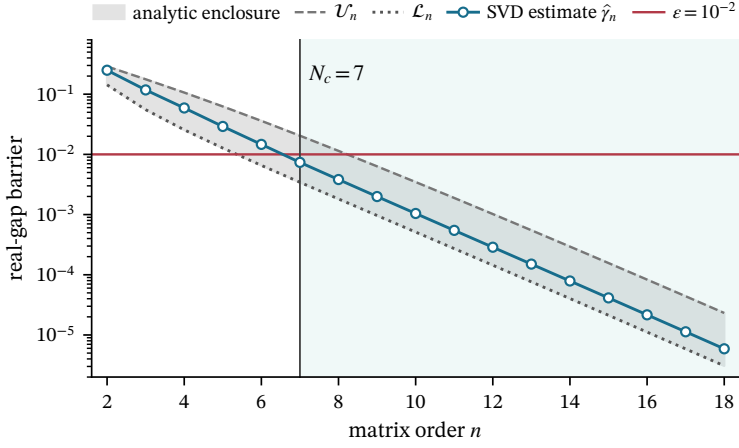


FIG. 2. Gapwise SVD estimates $\hat{\gamma}_n$ of the strictly decreasing barriers and the enclosure $\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a)$ for $a = 1/4$. The horizontal line $\varepsilon = 10^{-2}$ selects $N_c = 7$; the explicit bounds in (2.4) certify $6 \leq N_c \leq 9$. The nearly parallel log-scale curves display the common $(n+1)a^{n/2}$ scale.

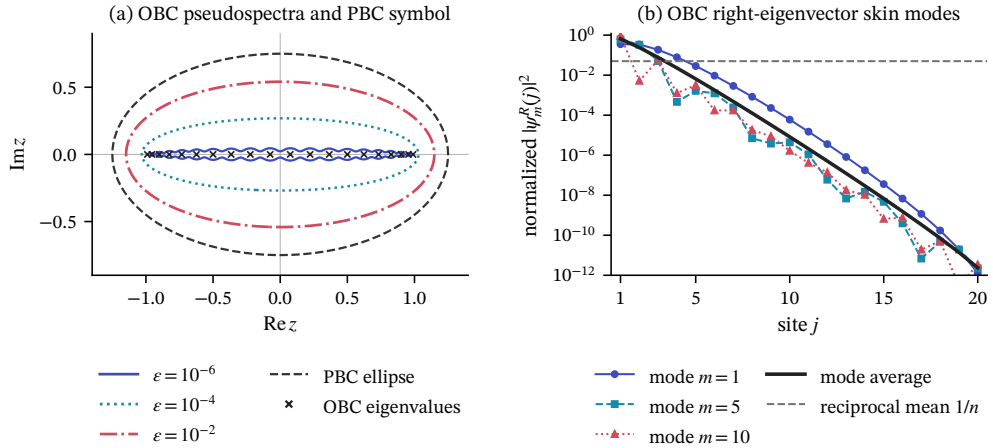


FIG. 3. Hatano–Nelson interpretation for $a = 1/4$ and $n = 20$. (a) The open-boundary eigenvalues lie in the real segment $[-2\sqrt{a}, 2\sqrt{a}]$, whereas the periodic symbol $e^{ik} + ae^{-ik}$ traces an ellipse. The nested contours are boundaries of strict pseudospectra. (b) Right eigenvectors $\psi_m^R(j) \propto a^{(j-1)/2} \sin \frac{m\pi j}{n+1}$ accumulate at the left edge. Reversing the site convention reverses the edge but not the pseudospectral conclusions.

7. Hatano–Nelson interpretation and conclusions. To connect the dimensionless matrix to the usual second-quantized notation, consider

$$\hat{H}_{\text{OBC}} = - \sum_{j=1}^{n-1} \left(t_L \hat{c}_j^\dagger \hat{c}_{j+1} + t_R \hat{c}_{j+1}^\dagger \hat{c}_j \right), \quad t_L > t_R > 0.$$

918 Its one-particle matrix is $-t_L A_n(a)$, $a = t_R/t_L$. The conventional minus sign is
 919 immaterial here because $\Sigma A_n(a) \Sigma = -A_n(a)$ for $\Sigma = \text{diag}(1, -1, \dots, (-1)^{n-1})$. Put

$$920 \quad t_0 = \sqrt{t_L t_R}, \quad h = \frac{1}{2} \log(t_L/t_R) = \frac{1}{2} \log(1/a).$$

921 For the unitarily equivalent positive-sign representative $t_L A_n(a)$, multiplying the
 922 dimensionless relation (1.4) by t_L gives the eigenvalues and unnormalized right and
 923 left modes

$$924 \quad E_m = 2t_0 \cos \frac{m\pi}{n+1},$$

$$925 \quad \psi_m^R(j) = e^{-h(j-1)} \sin \frac{j m \pi}{n+1},$$

$$926 \quad \psi_m^L(j) = e^{h(j-1)} \sin \frac{j m \pi}{n+1}, \quad 1 \leq j, m \leq n.$$

927 These mode profiles are closely related to the original Hatano–Nelson eigenfunction
 928 analysis [14]. Thus the right-mode amplitude skin depth is

$$929 \quad \ell_{\text{skin}} = h^{-1} = \frac{2}{\log(1/a)},$$

930 whereas the gauge condition number is $\kappa_2(\mathcal{G}_n(e^{-h})) = e^{h(n-1)} = a^{-(n-1)/2}$. The
 931 quantitative bounds in Proposition 5.2 give, more precisely,

$$932 \quad \gamma_n^{-1} \asymp_a \frac{e^{hn}}{n+1}.$$

933 Thus gauge conditioning and worst-case spectral sensitivity have the same exponential
 934 rate h , up to fixed- a constants and the displayed polynomial factor.

935 The perturbational characterization following (1.1) clarifies the meaning of connect-
 936 edness. It says that the *union* of eigenvalue locations attainable under all unstructured
 937 perturbations of norm below ε has merged into one uncertainty region; it does not
 938 identify a single perturbation whose eigenvalues populate all of the connecting bridges.
 939 Equivalently, for $A \in \mathbb{C}^{n \times n}$ and $z \notin \sigma(A)$,

$$940 \quad \|(zI_n - A)^{-1}\|_2 = \max_{\mathbf{f} \neq 0} \frac{\|\mathbf{u}\|_2}{\|\mathbf{f}\|_2}, \quad (zI_n - A)\mathbf{u} = \mathbf{f},$$

941 so the pseudospectrum is the region of worst-case resolvent gain larger than ε^{-1} ; at
 942 a spectral point the gain is interpreted as infinite. The quantity γ_n is the hardest
 943 real-axis bridge between adjacent open-boundary spectral islands.

944 For the dimensional Hamiltonian $t_L A_n(a)$, the merger level is $t_L \gamma_n$. If $\varepsilon_{\text{phys}}$ is a
 945 perturbation level with energy units and $\delta = \varepsilon_{\text{phys}}/t_L$ is its dimensionless strength,
 946 then (2.5) becomes

$$947 \quad N_c(a, \delta) = \ell_{\text{skin}} (\log(1/\delta) + \log \log(1/\delta)) + O_a(1).$$

948 The leading coefficient is exactly the amplitude skin depth. At fixed dimensionless δ ,
 949 its leading small- δ value decreases as a decreases; this is an asymptotic comparison,
 950 not a claim that N_c is monotone in a at every finite perturbation level. The limit
 951 $a \uparrow 1$ is not uniform: at $a = 1$ the matrix is normal and the connectedness scale is

governed by ordinary Hermitian eigenvalue spacing. Likewise, restricting perturbations to onsite, hopping, boundary, or Toeplitz structure can produce a smaller attainable set and requires a structured-pseudospectral analysis. Toeplitz-structured pseudospectra are treated in [7] and [10]. Robustness under random imperfections is studied in [3]; pseudospectral boundary sensitivity in correlated or disordered systems in [20]; and complex and structured-pseudospectral scaling in asymmetric lattices in [15].

In summary, the planar topology is controlled by a one-dimensional barrier sequence: vertical monotonicity makes every component contractible, persymmetry and folded Chebyshev coordinates make that sequence strictly decreasing with size, and rectangular factorizations identify its sharp scale. For the clean open Hatano–Nelson chain this provides an exact finite-size merger criterion, while keeping distinct the skin effect itself, worst-case spectral uncertainty, and perturbation-structured physical response.

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